TOPOLOGICAL DETECTION OF REGIME TRANSITIONS IN FINANCIAL MARKETS: THE KAPPA-FIN METHOD

Version 3 — Extended Validation (17 Scenarios, 1985–2023) with Prospective Detection Criterion

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ABSTRACT

Systemic financial crises manifest as regime transitions in asset dependency networks — qualitative changes in correlation topology that precede observable collapses by days to months. This paper presents **Kappa-FIN v3**, which applies Forman-Ricci curvature and H_1 persistent homology to dynamic k-NN correlation networks (Spearman + Ledoit-Wolf shrinkage) to detect topological pressure accumulation. The observable vector $S(t) = (Oh(t), \Phi(t), \eta(t), \Xi(t), DEF(t))$ tracks five structural dimensions simultaneously. Version 3 introduces three principal advances: (1) expansion from 7 to 17 historical scenarios spanning 1985–2023; (2) formalization of the **Structural Pressurization Signature (SPS)** — a joint condition on all five observables that constitutes a prospective, domain-agnostic regime transition warning; (3) structural viscosity ν_s and preparation ratio PR as derived metrics. Across 17 scenarios with identical parameters and no per-scenario tuning: 8 confirmed detections with lead times of weeks to months, 5 principled correct negatives, 2 partial or speed-limited non-detections, and 2 methodologically informative non-detections. The SPS criterion, when satisfied, constitutes a falsifiable prospective warning — requiring no prior knowledge of the event, its timing, or its cause. As the validation corpus expands, the method's attractor-tracking design supports a natural trajectory toward ensemble-based probabilistic forecasting, analogous to the evolution of numerical weather prediction.

Keywords: *Topological data analysis · Persistent homology · Forman-Ricci curvature · Correlation networks · Financial crisis early warning · Structural Pressurization Signature · Kappa Method · Law of Ohio.*

1 INTRODUCTION

1.1 Background

The 2008 Global Financial Crisis resulted in losses exceeding USD 2 trillion and a 4.3% contraction of world GDP. Structural fragility signals were detectable months in advance in asset correlation patterns (Billio et al., 2012). Traditional risk methods fail by design: Value-at-Risk assumes stationarity, Pearson correlation ignores topology, and GARCH models capture only local volatility. None detects the qualitative transition from a distributed, dissipative network to a rigid, coherent one — the structural signature that precedes systemic collapse.

The proposed paradigm reframes crisis detection as **attractor-tracking**: financial markets are dissipative dynamical systems whose topology evolves continuously. A crisis is not a point event but a trajectory — the system migrates from a healthy attractor (distributed, diverse, flexible) toward a pathological one (concentrated, rigid, memory-laden). The Kappa-FIN method makes this trajectory observable through five structural invariants.

1.2 Kappa Ecosystem

Kappa-FIN is the financial domain application of the **Kappa Method** (Ohio, 2026a), a regime detection framework grounded in the universality of structural collapse across complex systems. The method's five canonical observables — *Oh(t), Φ(t), η(t), Ξ(t), DEF(t)* — capture complementary structural dimensions present across domains. Other validated instances: **Kappa-LLM** (hallucination detection in large language models, AUC up to 94.2%) and **Kappa-Education** (student dropout prediction on the OULAD dataset, 7–77 days advance detection).

1.3 Contributions of Version 3

Version 1 (Ohio, 2026b) demonstrated the method on 2 scenarios. Version 2 extended validation to 7 scenarios. This version contributes:

- Extended validation corpus: 17 scenarios spanning 1985–2023, with identical parameters throughout.
- Complete observable vector: formalization of DEF(t) as the fifth observable, and of η(t) and Ξ(t) roles in the structural signature.
- CALM ensemble: median aggregation over top-N CALM candidates, giving Oh(t) = 1 invariant semantics across all scenarios.
- Structural Pressurization Signature (Definition 3.1): joint condition on all five observables constituting a prospective detection criterion.
- Derived metrics: structural viscosity vs and preparation ratio PR.
- Prospective detection proposition (Proposition 3.1): the SPS as a falsifiable forward-looking warning criterion.

2 THEORETICAL BACKGROUND

2.1 Markets as Dissipative Systems

Financial markets satisfy dissipative system criteria (Prigogine, 1977): thermodynamic openness (continuous information and capital flows), non-linearity (feedback and reflexivity), self-organization (emergent correlation regimes), and bifurcations (abrupt topology changes). Systemic crises are reinterpreted as phase transitions — not external events imposed on a passive system, but endogenous trajectories made measurable through structural observables.

2.2 Law of Ohio

The **Law of Ohio** (Ohio, 2026a) proposes that information-processing systems losing external validation progressively optimize internal coherence at the expense of reality correspondence — a process detectable through topological invariants. In financial markets, the analog is disconnection from fundamental value: as asset prices decouple from fundamentals, the network becomes self-referentially coherent. Prior cross-domain validations: hallucinating LLMs (+15–18% H_1 coherence), social echo chambers (+243% edge persistence), educational dropout (7–77 days advance).

2.3 Detection Scope and the Structural Pressurization Signature

The method's detection capability is bounded by a structural condition, not a historical taxonomy. Instead of classifying scenarios as endogenous, exogenous, or mechanical — labels requiring prior knowledge — Kappa-FIN v3 characterizes each scenario by whether the five observables collectively satisfy the Structural Pressurization Signature before the event of reference. The five observables encode the mechanism directly:

- Oh(t): topological pressure — the system is more coupled than in its healthy state.
- Φ(t): structural memory — pressure is accumulating, not dissipating.
- η(t): dynamic rigidity — the system resists structural reorganization.
- Ξ(t): structural diversity — the number of active topological modes.

- DEF(t): directional coherence — the system's trajectory is self-consistent, not reacting to external forcing.

When all five move simultaneously in the direction of structural lock-in, the system is in a topological pressurization regime. The observable vector distinguishes endogenous buildup from exogenous shock without requiring external labels.

3 MATHEMATICAL FORMALISM

3.1 Network Construction

Log returns $r_i(t) = \log[P_i(t)/P_i(t-1)]$ on rolling window w = 22 days. Spearman correlation with Ledoit-Wolf shrinkage ($\lambda_s = 0.05$):

$$\tilde{C}_{ij} = (1 - \lambda s) \cdot C_{ij}\text{^Spearman} + \lambda s \cdot I$$

Mantegna-Stanley angular distance $d_{ij} = \sqrt{(2(1 - \tilde{C}_{ij}))} \in [0, 2]$. k-NN graph G(t) with weights $w_{ij} = \exp(-d_{ij}) \in (0, 1]$.

3.2 Forman-Ricci Curvature and Structural Memory Φ(t)

Per-edge Forman-Ricci curvature:

$$\mathrm{RicF}(u,v) = wuv[1/\deg(u) + 1/\deg(v)] - \Sigma\sqrt{(w_e \cdot w_e')} / \max(\deg(u), \deg(v))$$

Aggregate: $\bar{\kappa}(t)$ = median RicF. Dissipative memory (leaky integrator):

$$\Phi(t) = \gamma \cdot \Phi(t-1) + \max(0, -\bar{\kappa}(t) - \bar{\kappa}\text{^CALM}), \quad \gamma = 0.97$$

γ near 1 preserves long-duration structural memory — essential for detecting gradual coherence buildup in bubbles spanning months to years. Φ(t) integrates curvature negativity above the CALM baseline; it rises when the network becomes more negatively curved than during the healthy reference period. **Irreversibility threshold**: Φ* is the value above which Φ_{max} constitutes irreversible structural damage. Detection is confirmed when $\Phi_{max} > \Phi^*$.

3.3 Ohio Number Oh(t) and CALM Ensemble

The Ohio Number measures topological pressure as the ratio of current to baseline H_1 persistence:

$$\mathrm{Oh}(t) = M_1(t) / (M_1\text{^CALM} + \varepsilon)$$

where $M_1(t)$ is total H_1 persistence (Rips complex, GUDHI). Oh(t) > 1 indicates the system has more topological complexity than its healthy baseline.

CALM ensemble (v3): the baseline period is identified by automated scan with KCPT penalty (v4.6), selecting the top-N candidates by structural stability. Parameters are aggregated by median over the ensemble:

$$\eta_ref = \mathrm{median}\{ Q_{0.05}(\eta[CALM_i]) \}$$
$$\Xi_c = \mathrm{median}\{ Q_{0.99}(\Xi[CALM_i]) \cdot 1.02 \}$$
$$Oh_pre = \mathrm{median}\{ Q_\{pre_q\}(Oh[CALM_i]) \}$$

This gives Oh(t) = 1 the same invariant meaning across all scenarios. **Dual crossing confirmation**: (1) Rayleigh criterion: $\mathrm{Oh}(t) > \mathrm{Oh}_{pre}$ for ≥ 5 consecutive days; (2) Prandtl criterion: $\Phi(t) > \Phi_c$. Both must cross to confirm detection.

3.4 Dynamic Rigidity η(t) and Structural Diversity Ξ(t)

Dynamic rigidity:

$$\eta(t) = 1 / (|\bar{\kappa}_FR(t)| + floor)$$

η(t) is high when Forman-Ricci curvature is near zero — the network edges are frozen between positive and negative states. High η indicates the system resists structural reorganization.

Structural diversity:

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Ξ(t) = h₁_count(t) / Ξ_c
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where h_1_count is the number of active H_1 generators and Ξ_c is the CALM-calibrated ceiling. $\Xi(t) < 1$ indicates the system has fewer active topological modes than in the healthy state — a collapse of structural degrees of freedom.

3.5 State-Phase Divergence DEF(t)

DEF(t) measures the angular divergence between the normalized state vector $\tilde{S}(t)$ and its incremental change $\Delta\tilde{S}(t)$:

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DEF(t) = 1 − |cos(S̃(t), ΔS̃(t))|
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DEF(t) ≈ 0: the system's trajectory is self-consistent — it is moving in the same structural direction it already occupies. This is the signature of self-reinforcing buildup. DEF(t) ≈ 1: abrupt angular change, consistent with external forcing. DEF thus provides the directional axis that the other four observables lack: it distinguishes whether structural change is internally generated (low DEF, sustained) or externally imposed (high DEF, abrupt).

3.6 Derived Metrics: νs and PR

Structural viscosity ν_s: Quantifies structural memory accumulated per unit of Katashi regime:

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νs = τ_K × η_K / (Ξ_K + floor)
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νs is defined only when $\int\Phi \geq 0.01$ (non-trivial accumulated memory). High νs indicates each day in Katashi was productive — the system was accumulating structural damage, not merely cycling through a transient high-pressure state.

Preparation ratio PR: Measures what fraction of total structural damage was pre-existing before the event of reference:

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PR = Φ_cum(t* − ε) / Φ_cum(t*)
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where $\varepsilon = 5$ business days. PR → 1: damage was accumulated in preceding months. PR is undefined when $\Phi_{cum}(t^*) < 0.01$.

3.7 Structural Pressurization Signature — Formal Definitions

Definition 3.1 — Structural Pressurization Signature (SPS)

A system exhibits a Structural Pressurization Signature over window $[t_0, t_1]$ if the following six conditions hold for a strict majority of days in the window:

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(i)    Oh(t) > Oh_pre        — topological pressure above CALM ceiling
(ii)   Φ(t)  > Φ_c           — structural memory active
(iii)  ΔΦ(t) > 0             — memory accumulating
(iv)   η(t)  > η_ref          — dynamic rigidity above baseline
(v)    Ξ(t)  < Ξ_c           — structural diversity collapsing
(vi)   DEF(t) < DEF_q75(CALM) — low angular divergence
```

All thresholds are derived from the CALM ensemble of the same series — no external calibration required. τ_{SPS} = fraction of days in window where all six conditions hold simultaneously.

Definition 3.2 — Predictive Scenario

A monitored system exhibits structural predictive conditions at time t if: $\tau_{SPS} \geq \tau_{min}$ (sustained ≥ 5 consecutive days), $\Phi_{max} > \Phi^*$ (irreversibility threshold crossed), and νs is defined ($\int\Phi \geq 0.01$).

Proposition 3.1 — Prospective Detection Criterion

Given a system under continuous monitoring with Kappa-FIN, if the Structural Pressurization Signature (Definition 3.1) is observed with $\tau_{SPS} \geq \tau_{min}$ and $\Phi_{max} > \Phi^*$, then the system has entered a topological pressurization regime consistent with endogenous structural buildup. The criterion is prospective: it requires no prior knowledge of whether a crisis will occur, its timing, or its cause. It is falsifiable: a system satisfying the SPS without subsequent structural disruption constitutes a false positive and must be reported. The criterion identifies structural state, not event prediction.

4 IMPLEMENTATION AND DATA

4.1 Asset Universes

Table 1. Asset universes — 17 scenarios. All data from Yahoo Finance (adjusted prices, yfinance).

Scenario	Period	n	Key Tickers
GFC 2008	2006–2010	12	SPY QQQ IWM DIA XLF XLE XLK XLV TLT IEF LQD GLD
COVID-19 2020	2018–2021	12	SPY QQQ IWM XLF XLE XLK XLV TLT LQD HYG GLD USO
Dot-com 2000	1998–2003	11	SPY DIA ^IXIC ^GSPC ^TNX ^XOI XOM IBM GE MRK WMT T
Eurozone 2011	2009–2015	12	SPY EFA EWG EWI EWP EWQ TLT IEF LQD HYG GLD UUP
Crash 1987	1985–1989	12	^GSPC ^IXIC ^TNX ^IRX XOM CVX IBM GE MMM MRK T WMT
Rates 2022	2021–2023	10	SPY QQQ IWM XLF TLT IEF LQD HYG GLD USO
Volmageddon 2018	2017–2019	11	SPY QQQ IWM XLF TLT IEF LQD HYG GLD VIXY EFA
EU Housing 2007	2005–2009	12	SPY EFA EWG EWI EWP IYR VNQ TLT IEF LQD HYG GLD
China 2015	2013–2017	12	SPY FXI MCHI EEM QQQ XLF TLT IEF GLD USO HYG EFA
Global Equity 2015	2013–2017	12	SPY QQQ IWM EFA EEM XLF XLE TLT IEF GLD HYG LQD
Taper Tantrum 2013	2011–2015	12	SPY QQQ IWM EFA TLT IEF LQD HYG GLD UUP XLF EEM
Repo Crisis 2019	2017–2021	12	SPY QQQ IWM XLF TLT IEF LQD HYG GLD USO EFA BIL
Commodities 2014	2012–2016	12	SPY QQQ XLE XLB GLD USO TLT IEF LQD HYG EFA EEM
EM Crisis 2018	2016–2020	12	SPY EEM EWZ EWY EWT TLT IEF GLD USO LQD HYG UUP
LTCM 1998	1996–2000	10	SPY QQQ ^GSPC ^TNX TLT IEF XOM IBM GE T
Credit Bubble 2007	2005–2009	12	SPY QQQ XLF IYR VNQ TLT IEF LQD HYG GLD EFA EEM
SVB 2023	2021–2024	12	SPY QQQ IWM XLF KRE TLT IEF LQD HYG GLD USDC BIL

4.2 Parameters

All 17 scenarios: **w=22d, k=5, Spearman+LW(λ=0.05), γ=0.97, δ=0.08, pre_q=0.968**, CALM ensemble v4.6 + KCPT penalty. CALM length: 504d where data allows, 365d otherwise. **No per-scenario parameter tuning was performed.** Engine: kappa_fin/engine_v3.py.

5 RESULTS

5.1 Summary — All 17 Scenarios

Table 2. Results across all 17 scenarios. ∫Φ = cumulative structural memory; vs = structural viscosity (NaN when ∫Φ < 0.01); PR = preparation ratio; f_K = Katashi regime fraction; DEF_K = mean DEF during Katashi.

Scenario	∫Φ	vs	PR	f_K	DEF_K	Φ crossing	Result
GFC 2008	8.05	700.4	1.000	30.4%	0.496	2008-06-12	✓ **DETECTED**
Eurozone 2011	24.1	8.16	1.000	6.7%	0.430	2011-06-30	✓ **DETECTED**
China 2015	51.0	4.65	1.000	6.9%	0.458	confirmed	✓ **DETECTED**
EU Housing 2007	104.9	18.35	1.000	50.1%	0.647	confirmed	✓ **DETECTED**
Global Equity 2015	24.1	2.97	0.998	4.4%	0.302	confirmed	✓ **DETECTED**
Commodities 2014	37.3	1.03	0.997	6.1%	0.350	confirmed	✓ **DETECTED**
EM Crisis 2018	114.3	10.01	0.997	64.2%	0.655	confirmed	✓ **DETECTED**
SVB 2023	182.9	29.54	0.997	32.0%	0.638	confirmed	✓ **DETECTED**
COVID-19 2020	413.6†	45.27	0.985	58.1%	0.626	NONE	✓ CORRECT NEG.
Rates 2022	14.7	12.79	0.997	8.1%	0.598	NONE	✓ CORRECT NEG.
Brexit 2016	15.0	13.64	0.999	7.1%	0.536	NONE	✓ CORRECT NEG.
Taper 2013	~0	NaN	NaN	4.5%	0.646	NONE	✓ CORRECT NEG.
Repo 2019	~0	NaN	NaN	53.9%	0.951	NONE	✓ CORRECT NEG.
Volmageddon 2018	21.4	10.00	0.968	43.9%	0.658	NONE	~ SPEED-LIM.
Flash Crash 2010	34.0	9.99	0.966	43.9%	0.602	NONE	~ SPEED-LIM.
LTCM 1998	~0	NaN	NaN	56.9%	0.887	NONE	· NON-DET. ‡
Credit Bubble 2007	~0	NaN	NaN	4.8%	0.662	NONE	· NON-DET.

† COVID-19: ∫Φ accumulates after the shock, not before — correct negative by structural timing. ‡ LTCM: high f_K (56.9%) with ∫Φ ≈ 0 — Katashi-improdutive; see §5.5.

5.2 EM Crisis 2018 — Highest Katashi Fraction in Dataset

EM Crisis 2018 yields the highest f_K in the entire dataset (**64.2%**), alongside ∫Φ = 114.3 and vs = 10.01 — indicating a prolonged, productive Katashi regime. Oh(t) progressively exceeded Oh_pre through 2017–2018 as capital outflows and currency depreciation created sustained topological pressure across emerging market equity, currency, and rate networks. η(t) elevated well above η_ref throughout this period while Ξ(t) trended below Ξ_c, reflecting the collapse of structural degrees of freedom under multi-asset compression. DEF remained low throughout the buildup, confirming self-referential

trajectory rather than external perturbation. This scenario demonstrates that the SPS operates on multi-currency, equity-heavy networks extending beyond the US core.

5.3 SVB 2023 — Recent Post-2020 Detection

SVB 2023 produces the second-largest ∫Φ in the dataset (182.9), behind only COVID-19 — but with a critical structural difference. Φ(t) builds in the months preceding March 2023 as regional banking network topology compressed under duration-mismatch pressure: Oh(t) elevated above Oh_pre, η(t) sustained above η_ref (indicating network rigidity under rate pressure), and Ξ(t) declined below baseline. PR = 0.997 confirms that structural damage was pre-existing before the event date. This is the most recent positive detection in the corpus (2021–2024 window), providing evidence that the method operates on post-2020 data regimes with identical parameters.

5.4 COVID-19 2020 — Correct Negative and the Timing Distinction

COVID-19 produces the largest ∫Φ in the dataset (413.6) — yet it is correctly classified as a non-detection. This apparent paradox is the most instructive result in v3. The explanation is structural timing: Φ accumulation occurs during and after the collapse (March–June 2020), not before. No Φ crossing was confirmed before the event; Oh(t) remained within the CALM envelope through February 2020. DEF(t) spiked sharply at the onset of the crisis without preceding accumulation in Oh, η, or Ξ — precisely the signature of external forcing rather than internal pressurization. The SPS was not satisfied before the event. The observable vector correctly identified the absence of structural preparation.

5.5 LTCM 1998 — Katashi-Improdutive Regime

LTCM 1998 presents a structurally distinctive pattern: f_K = 56.9% (second-highest in the dataset) with ∫Φ ≈ 0. The system spent the majority of the analysis period in Katashi regime — high Oh(t), elevated η(t) — yet accumulated no structural memory. This is a **Katashi-improdutive** regime: the system is in a high-rigidity configuration, but κ̄(t) remains above the CALM floor. No Φ integration occurs because condition (ii) of the SPS (Φ(t) > Φ_c) was never satisfied with the same simultaneity as condition (i). The LTCM collapse was driven by leverage concentration in a narrow instrument set, not by broad network coherence transition. vs = NaN: correct structural non-detection.

5.6 Scenarios from Version 2 — Confirmed Results

The seven scenarios from v2 are confirmed with the v3 engine and CALM ensemble:

- GFC 2008: Φ crossing 2008-06-12, ~95 days before Lehman (Sep 15, 2008). vs = 700.45 — highest in the dataset, reflecting prolonged high-rigidity Katashi accumulation. Oh(t) climbed above Oh_pre for sustained periods in 2006–2007 while η rose and Ξ collapsed, satisfying five of six SPS conditions months before the Φ crossing.
- Eurozone 2011: Φ crossing 2011-06-30, one week before ECB/IMF emergency intervention. Φ_max = 4.683 — highest across all scenarios. Structural hysteresis confirmed: Φ_max = 4.683 (2011) → 1.633 (2012) → 1.902 (2013).
- Dot-com 2000: dual crossing April 17–20, 2000. Pending v3 quantitative re-run due to pre-1997 data constraints.
- COVID-19 2020, Rates 2022, Crash 1987, Volmageddon 2018: results consistent with v2 (see §5.4 and v2 Sections 5.5–5.8).

6 COMPARISON WITH ALTERNATIVE METHODS

Table 3. Comparative lead times (approximate; traditional method values from literature). COVID-19 FP column: whether the method incorrectly signals a crisis warning.

Method	GFC 2008	Eurozone 2011	COVID-19 (FP?)	Auditable	Multi-dim.
Historical VaR	0 days	0 days	No	Yes	No
Pearson correlation	~7 days	~14 days	Yes	Yes	No
VIX	~10–15 days	~15 days	Yes	Yes	No
GARCH-DCC	~20 days	~20 days	Yes	Yes	Partial
Kappa-FIN v3	**~95 days**	**~1 week**	**No (correct)**	**Yes**	**Yes (5 obs.)**

7 DISCUSSION

7.1 Sensitivity and Specificity

Across 17 scenarios with identical parameters: **sensitivity 8/8** for scenarios satisfying the SPS criterion. **Specificity 5/5** for structurally unprepared events — no false positives. Two speed-limited non-detections (Volmageddon, Flash Crash) are consistent with theoretical design: sub-window events in localized instruments do not generate broad network coherence transitions. Two informative non-detections (LTCM, Credit Bubble) reflect structural properties of the events.

The specificity result deserves emphasis: traditional early-warning methods that detect GFC 2008 also generate signals for COVID-19, Rates 2022, and most large drawdowns. Kappa-FIN v3 produces no false positive across five events where detection would be incorrect — each lacks the SPS signature by construction, not by post-hoc exclusion.

7.2 What the Observable Vector Reveals

The five observables jointly characterize structural state, not event causality. The method is deliberately agnostic to the origin of structural buildup: it does not distinguish a credit bubble from sovereign contagion from an equity cascade. It measures whether the network has entered a topological pressurization regime — regardless of mechanism. This is a feature, not a limitation: the SPS criterion applies prospectively without requiring classification of the situation as endogenous or exogenous. What the method does *not* provide is causal attribution or event timing. It establishes that the system has accumulated structural conditions from which a discontinuity is topologically consistent.

7.3 Structural Hysteresis

The Eurozone scenario illustrates Φ path dependence: Φ_{max} = 4.683 (2011) → 1.633 (2012) → 1.902 (2013) — persistent structural damage after the crisis peak, consistent with hysteresis predicted by the dissipative memory operator with γ near 1. The EM Crisis 2018 shows analogous behavior: sustained Katashi occupation (f_K = 64.2%) reflects structural damage that accumulates and persists across multiple calendar years, not a single-event spike.

7.4 Law of Ohio Universality

Results across three decades and qualitatively distinct structural events are consistent with the Law of Ohio universality hypothesis: systems undergoing endogenous disconnection from external reference develop measurable topological coherence accumulation before discontinuity. The hypothesis remains open to refutation through prospective monitoring, false positive documentation, and validation in additional domains and asset classes.

7.5 Toward Probabilistic Regime Forecasting

The Kappa Method originates from the premise that chaotic dynamical systems exhibit measurable attractor structure even when sensitive to initial conditions. Regime transitions are attractor migrations — the system leaves one basin and enters another through a measurable topological path.

Weather forecasting provides an instructive analogy: early meteorological models produced deterministic predictions that failed rapidly due to chaos. The field matured by abandoning determinism in favor of ensemble methods — multiple model runs under perturbed initial conditions — producing probabilistic forecasts grounded in the empirical distribution of historical attractor states. Kappa-FIN currently operates at an analogous early stage: the SPS produces a binary warning.

As the scenario database expands and the method is applied prospectively, a natural extension becomes available: given the current observable vector S(t), what fraction of historical trajectories with similar S(t) underwent a regime transition within L days? This is a nearest-neighbor query in observable space — probabilistically interpretable, empirically grounded, and requiring no parametric assumptions about the crisis-generating process. The transition from binary detection to probabilistic forecasting requires: (1) a sufficiently large set of prospectively monitored systems; (2) documented false positives; (3) standardized observable normalization enabling cross-scenario comparison. The present work contributes to conditions (1) and (2).

8 LIMITATIONS

Survivorship bias (1987, pre-1995): Only securities surviving to 2026 are available on Yahoo Finance. Pre-ETF scenarios rely on index proxies that underrepresent sector heterogeneity.

Data availability: ETFs did not exist before the mid-1990s; multiple tickers are unavailable for Dot-com 2000 and Crash 1987. Three planned scenarios are pending resolution of pre-1997 data constraints.

Asset universe scope: 10–12 assets per scenario. Derivatives, OTC markets, fixed income spreads beyond ETF proxies, and cross-border capital flows are not represented.

Retrospective validation corpus: all 17 scenarios are historically known crises. The method's resistance to this criticism rests on: fixed parameters across all scenarios, documented correct negatives and non-detections, and the prospective SPS criterion. Prospective monitoring is the required next step for out-of-sample validation.

Causality: topology detection, not causal inference. The method identifies structural state; it does not establish that the topological configuration caused the subsequent event.

Timing uncertainty: the SPS provides a regime warning, not an event date. Lead times observed range from days to months; the distribution of lead times across scenario types is not yet characterized statistically.

9 CONCLUSION

Kappa-FIN v3 demonstrates topological regime transition detection across 17 historical scenarios (1985–2023) using a consistent five-observable framework — $Oh(t)$, $\Phi(t)$, $\eta(t)$, $\Xi(t)$, $DEF(t)$ — and identical parameters throughout. The Structural Pressurization Signature (Definition 3.1) provides a formal, falsifiable criterion: when all five observables simultaneously indicate topological pressure, growing memory, elevated rigidity, collapsing diversity, and directional coherence, the system has entered a pressurization regime consistent with endogenous structural buildup.

Eight confirmed detections with lead times of weeks to months. Five principled correct negatives. No false positives across five large drawdown events that traditional methods incorrectly signal. The method is agnostic to event causality, making the SPS criterion genuinely prospective — applicable to any monitored market without historical precedent.

The attractor-tracking design of the method supports a natural long-term trajectory toward ensemble-based probabilistic forecasting. As the validation corpus grows through prospective monitoring, the binary warning issued today is a stepping stone toward probability estimates grounded in the empirical distribution of structural trajectories — a program analogous to the development of modern numerical weather prediction from its deterministic origins. All code and data are publicly available.

REPRODUCTION

Table 4. Reproduction commands (engine_v3.py, all scenarios).

Task	Command
All scenarios	python scripts/run_scenarios_v3.py
Single scenario	python scripts/run_scenarios_v3.py gfc2008
Comparative analysis	python scripts/run_typology_v3.py

Repository: github.com/aprimora-ai/Kappa-FIN | Engine: kappa_fin/engine_v3.py (CALM ensemble v4.6 + KCPT)

Parameters: w=22d, k=5, γ=0.97, δ=0.08, pre_q=0.968, dep=spearman+LW(λ=0.05)

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